

Detailed Business Proposal

RegretGate - Action-aware pre-commit AI control plane | ControlPlane.ai

1. Problem framing

Generative AI already moves money, data, and people inside enterprises - customer chat, employee copilots, and regulated decision-support. Failures often surface only after the model has acted.

- Confidently wrong - hallucinations and poor decisions delivered with certainty
- Quietly expensive - token burn, tool thrash, runaway agent loops
- Subtly unsafe - bias, leakage, policy violations

Today's oversight is fragmented across quality evaluations, cost dashboards, and safety filters. Problems are detected after commit, when remediation is costly and liability is real.

Real-world complexities we design for

- Different use cases have different latency and risk budgets
- Bias, hallucination, and privacy risks often overlap
- No reliable real-time ground truth for every claim
- Over-flagging creates fatigue; under-flagging creates liability
- Multi-turn agents compound risk across downstream decisions
- Regulation differs by geography and industry
- Foundation models consumed via API - operate at input/output layer

Reference operating assumptions

- Multiple concurrent use cases (support, internal copilot, decision support)
- Tens of thousands of interactions per week combined
- Mix of well-governed and loosely-governed knowledge sources

2. Solution design

Expected Regret = $P(\text{failure}) \times \text{Impact} \times \text{Irreversibility}$

Core mechanism:

- Intent and action identification (reply / send / approve / refund / execute / tool loop)
- Parallel signal analysis: performance, cost/thrash, responsibility
- Regret estimation: calibrated 0-100 score, use-case and policy aware
- Responsibility override: hard block for leakage / safety / policy
- Intervention ladder: pass -> receipt -> soft rewrite -> hold/HITL -> hard block
- Receipts and audit trail for every decision
- Feedback loop: human outcomes recalibrate thresholds

Differentiation

- Typical checkers treat every output similarly; RegretGate is action-aware and regret-priced
- Typical tools are post-hoc; RegretGate is a pre-commit control plane
- Fixed rules age quickly; RegretGate uses policy packs + human feedback
- Quality / cost / safety are usually siloed; RegretGate unifies them

3. Target users

- AI Platform / MLOps lead - consistent gate without killing latency
- Risk / Compliance / Privacy - audit trail, geo packs, hard blocks

- Business process owners - HITL only where irreversibility is high
- CIO / Responsible AI sponsors - trust metrics and tunable tradeoffs

4. Business case and impact

- Latency preserved where safe (fast path for low regret)
- Liability reduced where irreversible (holds and hard blocks)
- Cost waste reduced via thrash detection
- Alert fatigue managed by allocating verification proportional to Expected Regret

Illustrative framing: 50,000 AI actions/week, ~3% high-regret, ~0.5% responsibility-critical. Even a small reduction in leakage events or erroneous automated payouts typically dominates checker operating cost.

5. Phased roadmap

- Phase 0 - Prototype (this submission): working PoC, scenarios, ops, docs
- Phase 1 - Pilot: shadow then enforce, real audit store, PII service, FP/FN labels
- Phase 2 - Platform: policy studio, compounding graph, thrash budgets, GRC export
- Phase 3 - Enterprise: multi-tenant controls, SDKs, continuous calibration

6. Key risks and mitigations

- Over-flagging -> ladder + tunable thresholds + fatigue metrics
- Under-flagging -> responsibility hard block independent of score
- No ground truth -> receipts, soft rewrite hedges, HITL holds
- Detector drift / aging rules -> feedback recalibration + versioned policy packs
- Latency regression -> parallel checks and fast path for low regret
- Bypass of gate -> platform embedding + audit incompleteness alerts

7. What Round 2 demonstrates

The prototype proves the core mechanism on sample data: score -> ladder -> hard block -> HITL -> feedback, across three use cases and three silent failure modes - without claiming production ML accuracy.

GitHub: <https://github.com/harshalDharpure/controlplane-regretgate>

Run: `npm install && npm run dev` -> `http://localhost:3000`